**Assignment 4**
**Deadline March 29**
**Peer Review deadline: April 1**

1. This question makes it clear that there is a difference between 'seeing' and 'observing' in physics. Synchronized clocks are stationed at regular intervals, a million km apart, along a straight line. When the clock next to you reads 12 noon:

(a) What time do you see on the 90th clock down the line?

(b) What time do you observe on that clock?

2. A rocket ship leaves earth at a speed of $\frac{3}{5}c$. When a clock on the rocket says 1 hour has elapsed, the rocket ship sends a light signal back to earth.

(a) According to earth clocks, when was the signal sent?

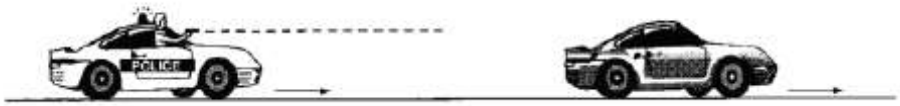
(b) According to earth clocks, how long after the rocket left did the signal arrive back on earth?

(c) According to the rocket observer, how long after the rocket left did the signal arrive back on earth?

3. A record turntable of radius R rotates at angular velocity ω. The circumference is presumably contracted, but the radius (being perpendicular to the velocity) is not. What is the ratio of the circumference to the diameter, in terms of ω and R? According to the rules of ordinary geometry, it has to be π. What's going on here?

4. Suppose a driver has a car and a garage of equal proper length. The driver speeds toward the garage and a doorman at the garage is instructed to slam the door shut as soon as the back end of the car enters the garage. According to the doorman, "the car Lorentz contracted and easily fitted into the garage when I slammed the door." According to the driver, "the garage Lorentz contracted and was too small for the car when I entered the garage." Explain what really happens? Is the doorman's statement correct? Is the driver's statement correct?

5. After committing a crime, a group of outlaws find a getaway car and try to escape. Due to overspeeding, the police are alerted and start to chase them. The outlaws' getaway car is moving at a speed of $\frac{3}{4}c$ (with respect to ground), while the police car is slow and is moving at $\frac{1}{2}c$ (with respect to ground). The police officer fires a bullet from his car to the getaway car with a velocity (relative to the gun) of $\frac{1}{3}c$.




Note that, there are four different observers/frames in this case, i.e., outlaws, police, bullet and ground. You have to calculate the velocity of each observer with respect to all other observers. From the information given above, fill the table given below. ('Do they escape' means did the bullet miss the outlaws for that observer)

| speed of → <br> relative to ↓ | Ground | Police | Outlaws | Bullet | Do they escape? |
|---|---|---|---|---|---|
| Ground | 0 | $\frac{1}{2}c$ | $\frac{3}{4}c$ | | |
| Police | | | | $\frac{1}{3}c$ | |
| Outlaws | | | | | |
| Bullet | | | | | |